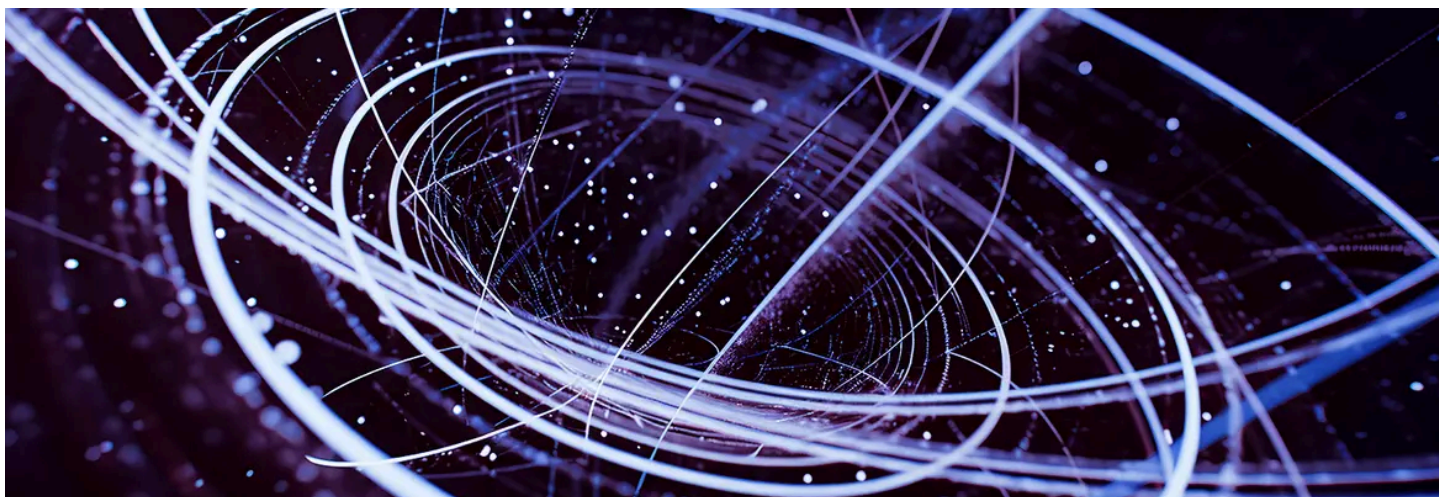


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New theory argues quantum physics must abandon irrational numbers and the continuum

The mysteries of quantum physics belong to maths, not nature



20th May 2026

 [cite](#)

Tim Palmer | Royal Society Research Professor Emeritus in the Department of Physics at the University of Oxford, and author of *The Primacy of Doubt*.

2,372 words

Read time: approx. 12 mins

*Quantum mechanics works perfectly in experiments, but Oxford physicist **Tim Palmer** argues it rests on a mathematical fiction: building the theory on the continuum of real numbers, including irrational numbers like $\sqrt{2}$. Palmer's radical alternative abandons the continuum of real numbers and eliminates quantum mysteries that aren't actually physical—from Schrödinger's cat, to Einstein's "spooky action at a distance." The theory makes a testable prediction: quantum computers will fail beyond 400 qubits, hitting a fundamental limit set by the discrete structure of nature itself. The coming quantum computing race will determine whether Palmer has identified a basic mistake in our most successful theory.*

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If you know one thing about quantum mechanics, the atomic theory formulated just over 100 years ago by Werner Heisenberg and Erwin Schrödinger, it is that quantum systems evolve by “jumping” discontinuously from one state to another; indeed, the words “quantum jump” and “quantum leap” have become part of everyday language.

If you know one other thing about quantum physics, it is that it describes a [deeply mysterious world](#) where cats can be simultaneously alive and dead and where choices about how we measure quantum systems on one side of the galaxy can instantaneously alter the states of quantum systems on the other side of the galaxy. [Such weirdness](#), we are sometimes told, is beyond the capability of the puny human brain to understand.

In this brief essay, I am going to try to persuade you that you are wrong about both of these apparent facts. On the first, I will argue that quantum mechanics depends much more vitally on the continuum of numbers than do the classical theories of physics that quantum mechanics replaced. Indeed, the iconic quantum jumps only occur at the endpoint of quantum evolution when we measure or observe a quantum system. Before then, quantum systems evolve absolutely continuously according to Schrödinger’s eponymous equation. On the second point, I argue below that if we “tweak” quantum mechanics by banishing the continuum in favour of a more discrete structure, all the so-called incomprehensible mysteries vanish in a puff of smoke. I refer to this theory, where the continuum has been banished, as [Rational Quantum Mechanics \(RaQM\)](#).

RaQM is, on the one hand, based on rational numbers, and on the other hand, its interpretation is totally comprehensible.

The existence of the mathematical continuum goes back to the fifth century BCE, when Hippasus of Metapontum showed that not all numbers can be described as the ratio of two whole numbers. Hippasus famously showed that the square root of 2 cannot be such a rational number. He was punished for having uncovered such a disturbing truth by being drowned at sea (either by his fellow Pythagorean followers or by the Gods, depending on which story you read). Today we have become rather blasé about the irrational numbers. However, I believe we have been lulled into a false sense of security.

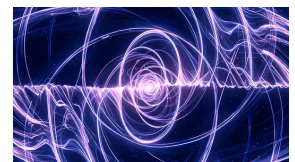
Until Newton and Leibniz invented the calculus, one could view Hippasus’s irrational numbers simply as an oddity. But the calculus required us to make sense of these irrational numbers not just here and there, but everywhere. Eighteenth and nineteenth-century mathematicians like Augustin-Louis Cauchy did their best to tame the irrational numbers, and in so doing made respectable the continuum field of real numbers, which combines the rational and irrational numbers into one set of numbers.



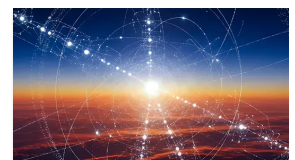
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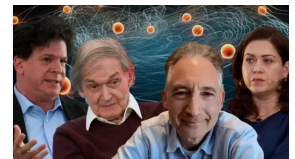
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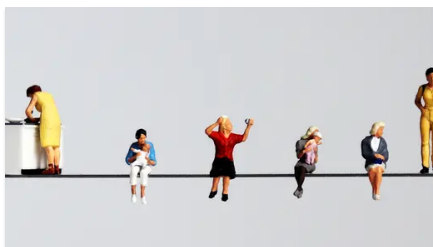


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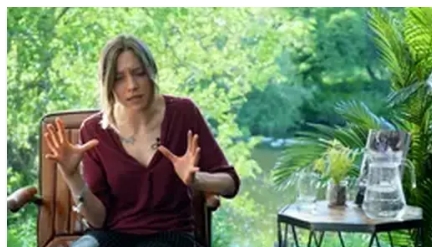


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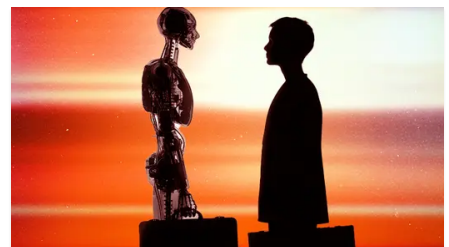
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